MIL-E-1/858A (USAF)
7 January 1972
SUPERSEDING
MIL-E-1/858 (USAF)
24 February 1955

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6542

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage regulator, subminiature

See Figure 1

Cathode --- Glow discharge

Base connections:

Pin No.	---	1	3	5
Element	---	k	a	k

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Total	Ez darkness	Ez Ambient light	Operating voltage	Operating current
Unit:		Vdc	Vdc	Vdc	mAdc
Maximum:		---	---	168	25
Minimum:		225	185	140	5
TEST CONDITIONS: (See note 1)		300	---	---	--

ABSOLUTE MAXIMUM RATINGS:

Parameter:	TA	TE	(A) Alt
Unit:	°C	°C	---
Maximum:	---	155	60,000
Minimum:	-55	---	---
TEST CONDITIONS: (See note 1)	25 +5	---	---

GENERAL:

Qualification - Required

Reliable Tube

(A) denotes changes

6542

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:45Z
Check the source to verify that this is the current version before use.

PAR. NO.	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 1 See note 1							
3347	Ionization voltage (1)	Ebb/Ib = 2 to 25 mAdc illumination = 5 to 50 foot-candles	0.65	II	Ez	---	180	Vdc
3337	Voltage drop (1)	Ebb/Ib = 25 mAdc	0.65	II	Etd	142	165	Vdc
3337	Voltage drop (2)	Ebb/Ib = 5 mAdc	0.65	II	Etd	142	165	Vdc
A 3335	Regulation	Etd(1) minus Etd(2)	0.65	II	Reg	---	6.0	Vdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2 See note 1							
1031	Variable-frequency vibration	No voltages; post-shock and vibration-fatigue test end points apply	10.0	See note 3	---	---	---	---
3347	Ionization voltage (2)	Ebb/Ib = 5 to 25 mAdc See note 4	6.5	Code F	Ez	---	225	Vdc
3305	Leakage current	Eb = 50 Vdc	6.5	Code F	Lib	---	20	µAdc
3278	Oscillation test	Esig = 15 mVac Ebb/Ib = 5 to 25 mAdc	2.5	1	---	---	---	---
3278	Noise test	Ebb/Ib = 25 mAdc	2.5	1	Eb	---	20	mVac
1116	Lead fatigue		2.5	Code F	---	---	---	---
2126	Glass strain		2.5	1	---	---	---	---
1041	Shock	450 G	---	---	---	---	---	---
1031	Vibration-fatigue	2.5G;fixed frequency F = 25 Hz (min), 60 Hz (max)	6.5	See note 3	---	---	---	---
----	Post-shock and vibration-fatigue test end points	Ionization voltage (1)	---	---	Ez	---	185	Vdc
		Regulation	---	---	Reg	---	8	Vdc
		Voltage drop (1)	---	---	Etd	---	140	Vdc
		Voltage drop (2)	---	---	Etd	---	140	Vdc
1105	Permanence of marking		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:45Z
Check the source to verify that this is the current version before use.

PAR. NO.	TEST	CONDITIONS	AQL (Percent Defective)	Insp Level or Code	Allowable Defectives Per Characteristic		Sym	LIMITS		Unit
					1st sample	Combined samples		Min	Max	
	Quality conformance inspection, part 3									
1521	Survival-rate life test	Intermittent life test, or equivalent conditions; TA=room	---	---	---	-------	---	---	---	---
----	Survival-rate life-test end point	Short and discontinuity detection	---	---	---	-------	---	---	---	---
1501	Intermittent life test	Ebb/Ib=25 mAdc TE =155°C (min) See notes 2, 5 and 6	---	---	---	-------	---	---	---	---
----	Intermittent life-test end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Regulation	---	---	2	4	Reg	---	8.0	Vdc
		Voltage drop (1)	---	---	2	4	Etd	140	168	Vdc
		Voltage drop (2)	---	---	2	4	Etd	140	168	Vdc
		Ionization voltage (1)	---	---	2	4	Ez	---	185	Vdc
		Total defectives	---	---	4	8	---	---	---	---
----	Intermittent life-test end points (1000 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Regulation	---	---	2	4	Reg	---	8.0	Vdc
		Voltage drop (1)	---	---	2	4	Etd	140	168	Vdc
		Voltage drop (2)	---	---	2	4	Etd	140	168	Vdc
		Ionization voltage (1)	---	---	2	4	Ez	---	185	Vdc
		Total defectives	---	---	4	8	---	---	---	---

NOTES:

1. A fixed resistor may be used and Ez varied to give the desired current.

2. Order for evaluation of life test defects: In the event a tube is defective for more than one attribute characteristic, the lowest numbered characteristic in this table shall constitute failure.

   1. Inoperative
   2. Regulation
   3. Tube Voltage Drop (1)
   4. Tube Voltage Drop (2)
   5. Ionization Voltage (1)

3. This test shall be conducted on the initial production lot and thereafter on a lot approximately every 30 days. In the event of lot failure, the lot shall be rejected and the succeeding lot shall be subjected to this test. Once a lot has passed, the 30 day rule shall apply. MIL-STD-105, sample size code letter F shall apply.

4. Conditions for this test shall be those of ionization voltage (1), except testing shall be done in total darkness and the tube shall not have conducted or have been exposed to light for at least 24 hours prior to testing.

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:45Z
Check the source to verify that this is the current version before use.

TES: -Continued

5. Bulb Temperature shall be defined as the highest temperature indicated when using a thermocouple of #40 BS or smaller diameter elements welded to a ring of .025 inch diameter phospher bronze placed in contact with the bulb.

6. Intermittent Life Tests:

a. Paragraph 4.6 of Specification MIL-E-1 shall be revised so that the mean electrode potentials (except heater and filament) shall not deviate by more than 5 percent from the specified values.

b. A life test defect is defined as a failure to meet the life test end point limits specified herein.

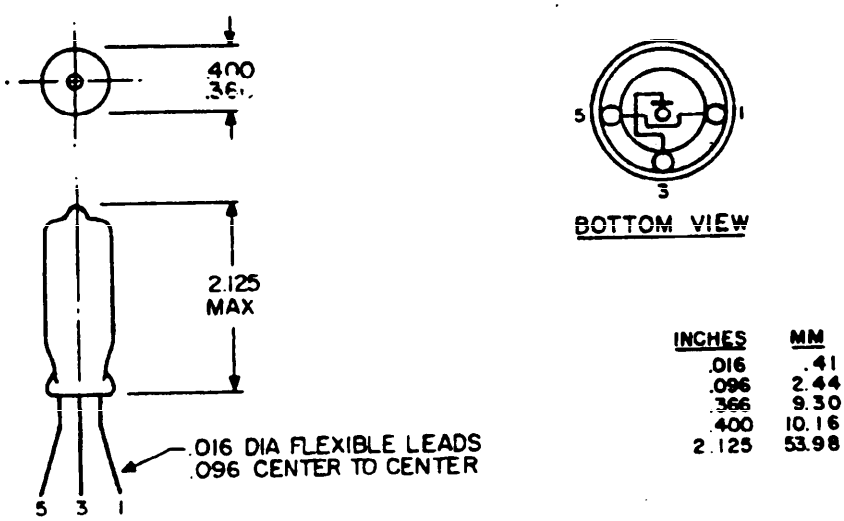
c. Lots shall not be released prior to completion of the 500 hour life test.

stodian:
Air Force - 80

Preparing activity:
Air Force - 80

(Project 5960-F898)

vlew activities:
Air Force 11, 17



INCHES	MM
.016	.41
.096	2.44
.366	9.30
.400	10.16
2.125	53.98

FIGURE 1 Outline dimensions of Electron Tube type 6542

NOTE:
1) Dimensions are in inches.
2) Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:45Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

5. Bulb Temperature shall be defined as the highest temperature indicated when using a thermocouple of #40 BS or smaller diameter elements welded to a ring of .025 inch diameter phosphor bronze placed in contact with the bulb.

6. Intermittent Life Tests:

   a. Paragraph 4.6 of Specification MIL-E-1 shall be revised so that the mean electrode potentials (except heater and filament) shall not deviate by more than 5 percent from the specified values.

   b. A life test defect is defined as a failure to meet the life test end point limits specified herein.

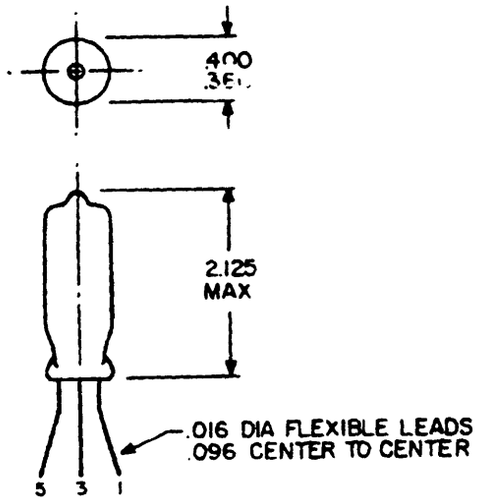
   c. Lots shall not be realeased prior to completion of the 500 hour life test.

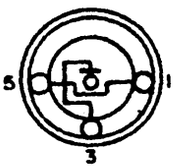
Custodian:
Air Force - 80

Review activities:
Air Force 11, 17

Preparing activity:
Air Force - 80

(Project 5960-F898)





BOTTOM VIEW

INCHES	MM
.016	.41
.096	2.44
.366	9.30
.400	10.16
2.125	53.98

FIGURE 1 Outline dimensions of Electron Tube type 6542

NOTE:
1) Dimensions are in inches.
2) Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:45Z
Check the source to verify that this is the current version before use.

OLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE

TED STATES AIR FORCE
ICIAL BUSINESS

SMAMA/MMREE
Wright-Patterson AFB OH 45433

OLD

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:45Z
Check the source to verify that this is the current version before use.